

Generating Structure-Consistent Hard Samples as Data Source for Robust Medical Image Segmentation

BY:

Tianjiao Wang

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Supervised By: Dr.Zhiqiang Gao

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AUTHOR: Tianjiao Wang

B.Sc., (Computer Science)

Wenzhou-Kean University

Wenzhou, China

SUPERVISORS: Dr. Zhiqiang Gao

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Abstract

Medical image segmentation models trained on limited annotated data frequently fail on challenging cases—small polyps, ambiguous boundaries, and visually confusing background textures. While synthetic data augmentation via generative models has shown promise, existing approaches generate samples that are model-specific and fail to generalize across different segmentation architectures. This thesis proposes a unified diffusion-based framework for generating structure-consistent hard samples that are universally challenging, regardless of the downstream model architecture.

The framework integrates four complementary components: (1) a BG-De local inpainting module that synthesizes morphologically distorted lesion regions while preserving global anatomical context; (2) an Attention Discrepancy Maximization (ADM) module that enforces strict geometric alignment between generated lesions and ROI masks; (3) a Prototype-level Feature Shift (PFS) module that constructs hard positives and confusing false negatives by manipulating deep feature representations relative to real-data prototypes; and (4) a DADA-style adversarial perturbation module that injects structured pseudo-lesion noise into background regions to simulate realistic false-positive scenarios.

Experiments are conducted on five publicly available polyp segmentation benchmarks. The diffusion model is fine-tuned on the full 1,450-image training corpus; during downstream evaluation, an additional 1,450 synthetic images are generated and combined with real images to train three segmentation backbones—PraNet, Polyp-PVT, and U-Net. Ablation studies compare four training configurations—Baseline (Real 1450), Real + G_easy, Real + G_hard1, and Real + G_hard2 (Ours)—across five benchmark datasets. The full model achieves an ETIS mDice of 0.7189 (+3.0 pp over baseline) and demonstrates that DADA is the critical component for cross-domain generalisation. Visual comparisons further confirm the structural fidelity and clinical plausibility of generated hard samples.

Generation quality is validated through FID, LPIPS, and IS comparisons against LaMa, RePaint, DetDiffusion, and HNS. Ablation studies compare four training configurations—Baseline (Real 1450), Real + G_easy (O), Real + G_hard1, and Real + G_hard2 (Ours)—across five benchmark datasets.

Keywords: Diffusion Models; Hard Sample Generation; Medical Image Segmentation; Polyp Detection; Data Augmentation.

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Definition of Acronyms

Acronym	Definition
DDPM	Denoising Diffusion Probabilistic Model
LDM	Latent Diffusion Model
SD	Stable Diffusion
LoRA	Low-Rank Adaptation
ADM	Attention Discrepancy Maximization
BG-De	Background Decoupling
PFS	Prototype-level Feature Shift
DADA	Detector-guided Adversarial Diffusion Attacker
ROI	Region of Interest
FID	Fréchet Inception Distance
LPIPS	Learned Perceptual Image Patch Similarity
IS	Inception Score
mIoU	Mean Intersection over Union
MAE	Mean Absolute Error
mDice	Mean Dice Coefficient

Chapter 1

Introduction

1.1 Overview of Medical Image Synthesis

Medical image synthesis has become increasingly essential in modern clinical AI systems, particularly in scenarios where data scarcity, expensive manual annotations, and strict patient privacy constraints severely limit the availability of high-quality training data. Colorectal cancer remains one of the most prevalent and lethal cancers worldwide, with colonoscopy-based polyp segmentation serving as the primary screening and early-detection mechanism. Despite significant advances in deep learning-based segmentation models, their clinical reliability is constrained by the limited diversity and volume of annotated training data.

By generating realistic and clinically plausible synthetic images, generative models can mitigate data scarcity while simultaneously introducing structured difficulty into training pipelines. High-fidelity synthetic data can reproduce complex lesion

morphology, fine-grained texture patterns, and anatomical structures that rarely appear in standard training sets. This dual capability—augmentation and difficulty modulation—has driven a paradigm shift from traditional data augmentation toward hard sample generation, which seeks to produce not only realistic samples but also semantically and structurally challenging cases.

1.2 Motivation

The rapid development of deep learning and generative artificial intelligence has significantly advanced medical image analysis, particularly in semantic segmentation and computer-aided diagnosis. Early generative paradigms such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) demonstrated the potential of realistic medical image synthesis, while more recent transformer-based architectures improved global feature modeling and structural representation capabilities.

In colorectal cancer screening, colonoscopy-based polyp segmentation plays an essential role in early detection and clinical decision-making. However, modern segmentation models still rely heavily on large-scale annotated datasets, while medical image collection is constrained by limited data availability, expensive expert annotations, and strict patient privacy regulations. Recent advances in diffusion models and latent diffusion frameworks have further shifted research attention from traditional augmentation toward controllable hard sample generation for improving model robustness.

This research was motivated by the increasing need for segmentation models that can generalize reliably under realistic clinical conditions. In practical colonoscopy scenarios, models frequently encounter difficult cases such as flat polyps with weak boundaries, tiny lesions hidden in mucosal folds, and visually confusing background structures. Existing augmentation methods mainly focus on visual realism and often fail to generate semantically difficult yet anatomically consistent samples.

Recent studies on diffusion-guided synthesis and hard negative generation have demonstrated promising results for improving segmentation robustness under domain shift. Therefore, this work proposes a structure-consistent hard-sample generation framework designed to improve downstream segmentation robustness, cross-dataset generalization, and false-positive resistance in medical image analysis.

1.3 Problem Statement

Despite the promise of diffusion-based synthesis, existing methods for hard sample generation suffer from a fundamental limitation: model specificity. Samples generated using feedback from a specific downstream segmentation model reflect that model’s unique attention patterns and decision boundaries, rendering them challenging only for that particular architecture. Different segmentation models—UNet, PraNet, Polyp-PVT—rely on distinct discriminative cues: some emphasize boundary gradients, others focus on texture homogeneity within lesion regions, and others model long-range contextual relationships.

Furthermore, feedback-driven generation pipelines are prone to semantic misalignment, geometric drift, and hallucinated pseudo-structures that compromise anatomical plausibility. These issues collectively undermine the practical value of existing hard sample generation methods in real clinical workflows.

1.4 Proposed Framework Overview

This thesis proposes a unified, model-agnostic hard sample generation framework that integrates structural manipulation, attention-guided geometric alignment, semantic feature perturbation, and adversarial background synthesis into a single coherent diffusion pipeline. The framework operates without requiring gradients from downstream task models, ensuring that generated samples are universally challenging across diverse segmentation architectures.

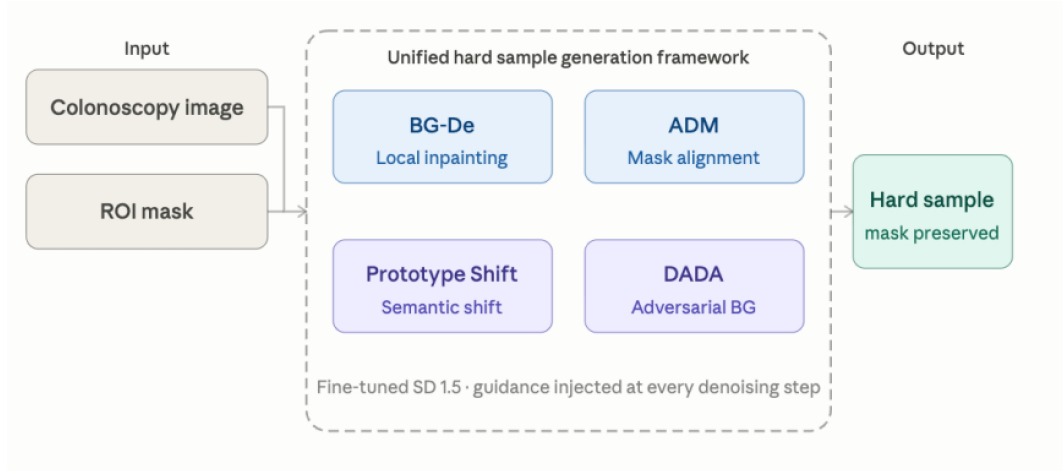


Figure 1.1: Overview of the proposed unified hard sample generation framework

Figure 1.1 illustrates the high-level architecture of the proposed framework. The pipeline accepts a colonoscopy image and its ROI mask as input, applies four com-

plementary modules (BG-De, ADM, Prototype Shift, DADA), and outputs a hard sample retaining the original mask annotation. Unlike prior approaches that operate under a constrained low-data regime, our framework is fine-tuned on the full 1,450-image training corpus and generates an equal number of synthetic hard samples for downstream evaluation.

1.5 Main Contributions

- A model-agnostic hard sample generation framework that integrates local structural manipulation, attention-geometric alignment, semantic prototype shifting, and pseudo-structure perturbation into a single diffusion pipeline, without requiring downstream model gradients.
- The BG-De inpainting module, which introduces controllable morphological distortion, boundary ambiguity, and texture irregularities inside the ROI mask while preserving global anatomical integrity.
- The Attention Discrepancy Maximization (ADM) module, which uses cross-attention maps as alignment signals at every denoising step to prevent label drift and maintain strict ROI-mask consistency.
- The Prototype-level Feature Shift (PFS) module, which constructs hard positives and confusing false negatives by perturbing deep ROI feature representations relative to real-data class prototypes.
- A DADA-style adversarial perturbation module that injects structured pseudo-

lesion noise guided by ADM residuals into background regions, generating clinically realistic false-positive artifacts.

- A comprehensive experimental evaluation on five polyp segmentation benchmarks, including generation quality comparison (FID, LPIPS, IS) against four state-of-the-art generation methods, visual comparisons of generated outputs, and ablation studies across four training configurations, demonstrating consistent improvements across three segmentation backbones.

1.6 Objectives and Scope

The primary objective of this thesis is to develop a unified diffusion-based hard sample generation framework for robust medical image segmentation. The proposed framework aims to generate structure-consistent and semantically challenging synthetic samples that improve downstream segmentation robustness across multiple architectures, including U-Net, PraNet, and Polyp-PVT.

More specifically, this research aims to:

- Design a model-agnostic hard sample generation pipeline that does not rely on downstream model gradients.
- Introduce structural perturbation, semantic feature manipulation, geometric alignment, and adversarial background synthesis into a unified diffusion framework.
- Improve cross-dataset generalization and false-positive robustness for medical

image segmentation.

- Evaluate the effectiveness of the proposed framework using multiple public polyp segmentation benchmarks and generation-quality metrics.

The scope of this research is limited to 2D colonoscopy image segmentation and diffusion-based synthetic image generation. The study focuses specifically on polyp segmentation datasets, including Kvasir-SEG, CVC-ClinicDB, ETIS, ColonDB, and EndoScene. The proposed framework is evaluated using downstream segmentation metrics such as mDice and mIoU, as well as generation-quality metrics including FID, LPIPS, and IS.

This thesis does not explore: 3D medical image synthesis for CT or MRI volumes; Real-time clinical deployment; Reinforcement learning or multimodal language-image generation; Large-scale clinical trials or physician-based validation studies.

1.7 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews related work across generative modeling, medical image synthesis, and hard sample generation. Chapter 3 introduces the proposed methodology, including BG-De, ADM, PFS, and DADA modules. Chapter 4 describes experimental settings, benchmark datasets, ablation studies, and qualitative analysis results. Chapter 5 concludes the thesis and outlines possible future research directions.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the literature related to generative medical image synthesis, hard sample generation, and robust medical image segmentation. The review focuses on three major research directions: generative models in medical imaging, diffusion-based medical image augmentation, and hard sample generation with false-positive modeling. Existing methods and their limitations are analyzed to motivate the framework proposed in this thesis.

2.2 Previous Work

2.2.1 Generative Models in Medical Imaging

Generative models have become an important research direction in medical image analysis due to their ability to synthesize realistic training data and alleviate annotation scarcity. Early approaches mainly relied on Generative Adversarial Networks (GANs), which achieved high perceptual realism through adversarial training¹. Variants such as StyleGAN² and conditional GANs have been widely applied to CT, MRI, and endoscopic image synthesis. However, GAN-based methods often suffer from training instability, mode collapse, and limited controllability when generating complex lesion structures.

Recent advances in Denoising Diffusion Probabilistic Models (DDPMs)³ and Latent Diffusion Models (LDMs)⁴ have significantly improved generation quality and stability. Diffusion models iteratively denoise Gaussian noise to synthesize high-fidelity images while preserving structural consistency⁵. Stable Diffusion further reduces computational overhead by operating in latent space⁶ and supports efficient domain adaptation through Low-Rank Adaptation (LoRA)⁷. These characteristics make diffusion models particularly suitable for controllable medical image synthesis.

Several studies have extended diffusion models to medical imaging tasks. Polyp-LDM⁸ introduced a shared VAE architecture⁹ to improve boundary alignment between generated polyp regions and segmentation masks, while Diff-VPS¹⁰ explored

diffusion-based temporal consistency for video polyp segmentation. These methods demonstrate the potential of diffusion models in medical image generation but still mainly focus on perceptual realism rather than downstream robustness improvement.

2.2.2 Medical Image Synthesis for Data Augmentation

Medical image synthesis has been widely used to address limited training data and expensive annotation costs. Traditional augmentation methods rely on geometric transformations, lesion insertion, or image inpainting to increase data diversity. Pseudo-lesion insertion approaches paste lesion crops into normal backgrounds, while inpainting methods such as LaMa¹¹ and RePaint¹² synthesize missing regions through context-aware restoration.

More recent methods incorporate structural priors and Transformer-based architectures to improve anatomical consistency. Structure-aware diffusion models¹³ enforce geometric constraints during synthesis, and cross-modal Transformer frameworks¹⁴ enhance long-range dependency modeling for lesion generation. Although these approaches improve visual realism, most methods optimize only for image quality and fail to generate semantically challenging samples for downstream segmentation models.

The ASPS framework¹⁵ further demonstrated that targeted structural perturbation can improve polyp segmentation robustness by enhancing feature diversity. However, existing methods still lack a unified mechanism for simultaneously controlling geometry, semantic difficulty, and false-positive structures.

2.2.3 Hard Sample Generation and False-Positive Modeling

Hard sample generation aims to construct training samples that expose weaknesses of downstream models. Existing methods generally focus on adversarial synthesis or feature-space perturbation.

Detector-guided adversarial diffusion methods, such as DetDiffusion¹⁶, use gradients from downstream segmentation or detection models to guide sample generation toward difficult cases. While effective for a single target model, these approaches produce architecture-specific hard samples and show limited transferability across different segmentation backbones¹⁷.

Hard Negative Synthesis (HNS)¹⁸ perturbs semantic feature representations in embedding space to move samples closer to decision boundaries. Although this improves semantic difficulty, it lacks explicit geometric control and cannot generate anatomically realistic false-positive structures in image space.

Other studies explored adversarial background perturbation¹⁹ to simulate clinically realistic false-positive artifacts such as vascular textures, specular highlights, and mucosal folds. These approaches demonstrated that explicitly modeling confusing background structures can improve robustness in real colonoscopy environments²⁰.

Inspired by these studies, this thesis integrates multiple guidance mechanisms into a unified diffusion framework. The proposed method combines:

- BG-De for local lesion structure perturbation,

- ADM for attention-guided ROI alignment,
- Prototype Feature Shift (PFS) for semantic feature manipulation,
- DADA for adversarial false-positive background synthesis.

The framework generates structurally consistent yet challenging medical training samples with blurred boundaries, texture ambiguity, and pseudo-lesion interference while preserving original ROI annotations.

2.3 Research Gaps

Although existing research has achieved significant progress in medical image synthesis and hard sample generation, several limitations remain:

- Most diffusion-based synthesis methods[?] primarily optimize perceptual realism rather than downstream segmentation robustness.
- Existing hard sample generation approaches²² rely heavily on downstream model gradients, producing architecture-specific samples that generalize poorly across different segmentation models.
- Feature-space perturbation methods²³ improve semantic difficulty but often lack geometric consistency and annotation reliability.
- Existing false-positive synthesis approaches²⁴ cannot simultaneously preserve ROI-mask alignment while generating clinically realistic background interference.

- No existing framework jointly integrates structural perturbation²⁵, semantic feature manipulation²⁶, attention-guided alignment, and adversarial background synthesis into a unified model-agnostic pipeline.

To address these limitations, this thesis proposes a unified diffusion-based hard sample generation framework for robust medical image segmentation.

2.4 Summary

This chapter reviewed the existing literature on generative medical imaging, diffusion-based image augmentation, and hard sample generation. Previous studies demonstrated the effectiveness of diffusion models for realistic medical image synthesis and showed that hard sample generation can improve downstream robustness. However, current methods still suffer from limitations in geometric consistency, semantic controllability, and cross-model generalization.

Motivated by these research gaps, this thesis proposes a unified framework integrating lesion structure perturbation, attention-guided alignment, semantic feature shifting, and adversarial false-positive synthesis. The next chapter introduces the formal problem formulation and the overall framework design in detail.

Chapter 3

Methodology

3.1 Problem Formulation

This section formalizes the hard sample generation task as a constrained conditional generation problem. Given an input medical image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ and its corresponding binary ROI mask $M \in \{0, 1\}^{H \times W}$, we define a hard sample generation function $G_\theta : (\mathbf{x}, M) \rightarrow \hat{\mathbf{x}}$, where $\hat{\mathbf{x}}$ denotes the synthesized hard sample. The function G_θ must satisfy three simultaneous constraints:

- **Geometric Alignment:** the synthesized ROI region $\hat{\mathbf{x}}_{\text{ROI}} = \hat{\mathbf{x}} \odot M$ must remain geometrically consistent with the masked lesion location.
- **Semantic Difficulty:** the deep feature representation $\phi(\hat{\mathbf{x}} \odot M)$ of the generated ROI should be controllably shifted relative to the real-data prototype $\mathbf{p}$.
- **False-Positive Controllability:** the background region $\hat{\mathbf{x}} \odot (1 - M)$ should

contain realistic yet non-pathological pseudo-structures that simulate the appearance of polyp-like artifacts.

3.2 Framework Design

3.2.1 Overall Framework

The proposed framework conditions a fine-tuned Stable Diffusion 1.5 model on ROI masks to perform localized hard sample synthesis. The overall pipeline processes an input image-mask pair through four sequential yet jointly optimized modules: BG-De inpainting for structural manipulation, ADM for geometric alignment, Prototype-level Feature Shift for semantic difficulty, and DADA-style adversarial perturbation for false-positive background synthesis.

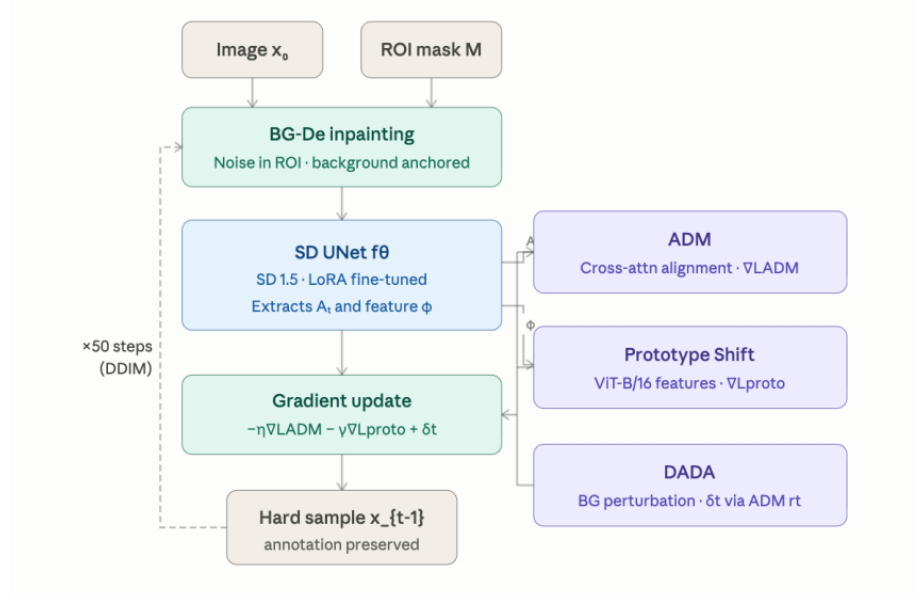


Figure 3.1: Detailed architecture diagram of the proposed unified hard sample generation framework

3.2.2 Diffusion Framework

Our approach builds upon the DDPM framework, in which a forward process q progressively adds Gaussian noise to a clean image $\mathbf{x}_0$ over T timesteps:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}\right) \quad (3.1)$$

The learned reverse process p_θ recovers $\mathbf{x}_0$ from $\mathbf{x}_T$ via iterative denoising:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)) \quad (3.2)$$

Our hard sample generation framework modifies the standard reverse process by injecting four complementary guidance signals at each denoising step, conditioning the trajectory on both the ROI mask M and the difficulty objectives defined above.

3.2.3 Evaluation Framework

We evaluate the proposed framework from two complementary perspectives. For generation quality, we adopt FID to measure distribution-level similarity between generated and real images (lower is better); LPIPS to assess perceptual diversity and local structural variation (higher indicates greater diversity); and IS to evaluate semantic distinguishability. For downstream segmentation performance, we use mDice, mIoU, MAE, and average score across five benchmark datasets.

3.3 Algorithm: Hard Sample Generation

The following algorithm presents the complete step-by-step procedure for generating structure-consistent hard samples. All mathematical equations used in the framework are defined below and referenced throughout [Algorithm 3.1](#).

3.3.1 Module 1: Background Decoupling Inpainting (BG-De)

The BG-De module treats the ROI mask M as a spatial constraint and performs controlled structural manipulation within the masked region at every denoising timestep. We construct a locally editable latent by blending random noise within the ROI with the original noisy image outside it:

$$\tilde{\mathbf{x}}_t = M \odot \mathbf{z}_t + (1 - M) \odot \mathbf{x}_t, \quad \mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3.3)$$

The diffusion model then predicts $\mathbf{x}_{t-1} = f_\theta(\tilde{\mathbf{x}}_t, t)$, generating localized structural variations exclusively within the ROI while the background $\mathbf{x}_t \odot (1 - M)$ remains anchored to the original image distribution. This introduces controllable morphological distortion — boundary blurring, shape deformation, and texture irregularities — corresponding to clinically observed edge-ambiguous polyp presentations.

3.3.2 Module 2: Attention Discrepancy Maximization (ADM)

Label drift — the misalignment between generated lesion content and the conditioning mask — is a well-documented failure mode in mask-conditioned diffusion models. The

Algorithm 3.1 Unified Hard Sample Generation Framework

Input: Image $\mathbf{x}$, ROI mask M , diffusion model f_θ , hyperparameters $\eta, \gamma, \varepsilon, K, T$

Output: Hard sample $\hat{\mathbf{x}}$

// **Phase 1: LoRA Fine-tuning**

Initialize f_θ from Stable Diffusion v1.5 pre-trained weights[?]

Fine-tune f_θ with LoRA[?] using rank $r = 64$, $\alpha = 32$, 3,000 steps, peak LR = 2×10^{-4} , FP16 mixed precision

// **Phase 2: Hard Sample Inference (DDIM, $T = 50$)**

Sample initial noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

for $t = T, T-1, \dots, 1$ **do**

[BG-De] Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; construct blended latent:

$$\tilde{\mathbf{x}}_t = M \odot \mathbf{z}_t + (1 - M) \odot \mathbf{x}_t$$

[ADM] Extract cross-attention map from UNet decoder:

$$A_t = \text{Softmax}\left(\frac{Q_t K_t^\top}{\sqrt{d}}\right) V_t$$

 Compute ADM loss $\mathcal{L}_{\text{ADM}} = \mathcal{L}_{\text{Fl}} + \mathcal{L}_{\text{Fe}}$, where:

$$\mathcal{L}_{\text{Fl}} = - \sum_{(i,j) \in M} A_t(i,j) + \sum_{(i,j) \notin M} A_t(i,j), \quad \mathcal{L}_{\text{Fe}} = \sum_{k \in \text{topK}(A_t \odot (1-M))} A_t(k)$$

[PFS] Extract ROI features $\phi(\tilde{\mathbf{x}}_t, M)$ via frozen ViT-B/16[?]; compute prototype loss:

$$\mathcal{L}_{\text{proto}} = \begin{cases} + \|\phi(\tilde{\mathbf{x}}_t, M) - \mathbf{p}\|^2 & \text{(hard positive)} \\ - \|\phi(\tilde{\mathbf{x}}_t, M) - \mathbf{p}\|^2 & \text{(confusing negative)} \end{cases}$$

 Anneal γ from high to low over sampling trajectory

[DADA] Derive background perturbation from ADM residual:

$$\mathbf{r}_t = \nabla_{\mathbf{x}_t}(\mathcal{L}_{\text{Fl}} + \mathcal{L}_{\text{Fe}}), \quad \delta_t = \varepsilon \cdot \text{sign}(\mathbf{r}_t) \odot (1 - M)$$

[Update] Apply joint denoising step:

$$\mathbf{x}_{t-1} = f_\theta(\tilde{\mathbf{x}}_t, t) - \eta \nabla_{\mathbf{x}_t} \mathcal{L}_{\text{ADM}} - \gamma \nabla_{\mathbf{x}_t} \mathcal{L}_{\text{proto}} + \delta_t$$

end for

return $\hat{\mathbf{x}} = \mathbf{x}_0$

(generated at 512×512, resized to 352×352)

ADM module addresses this by using cross-attention maps extracted from the UNet decoder as continuous geometric alignment signals:

$$A_t = \text{Softmax}\left(\frac{Q_t K_t^\top}{\sqrt{d}}\right) V_t \quad (3.4)$$

The ADM loss combines an attention contrast term and a background suppression term:

$$\mathcal{L}_{\text{Fl}} = - \sum_{(i,j) \in M} A_t(i,j) + \sum_{(i,j) \notin M} A_t(i,j) \quad (3.5a)$$

$$\mathcal{L}_{\text{Fe}} = \sum_{k \in \text{topK}(A_t \odot (1-M))} A_t(k) \quad (3.5b)$$

$$\mathcal{L}_{\text{ADM}} = \mathcal{L}_{\text{Fl}} + \mathcal{L}_{\text{Fe}} \quad (3.6)$$

The ADM gradient is injected into the sampling update at each step (see [Equation 3.9](#)), where η is the step-size hyperparameter. This maintains strict geometric alignment between generated lesion regions and input masks throughout the entire generative trajectory, enabling safe reuse of original mask annotations without re-annotation.

3.3.3 Module 3: Prototype-Level Feature Shift (PFS)

To control the semantic difficulty of generated samples, the PFS module manipulates deep feature representations extracted from intermediate synthesis states. At each step t , ROI features $\phi(\tilde{\mathbf{x}}_t, M)$ are extracted via a frozen ViT-B/16 backbone[?] and

compared against the class prototype $\mathbf{p}$, defined as the mean ROI feature embedding across the real training set. Two complementary objectives implement controllable difficulty:

$$\mathcal{L}_{\text{proto}} = \|\phi(\tilde{\mathbf{x}}_t, M) - \mathbf{p}\|^2 \quad (\text{hard positive}) \quad (3.7a)$$

$$\mathcal{L}_{\text{proto}} = -\|\phi(\tilde{\mathbf{x}}_t, M) - \mathbf{p}\|^2 \quad (\text{confusing negative}) \quad (3.7b)$$

The hard positive objective pushes generated ROI features away from the real lesion prototype, producing samples with exaggerated characteristics; the confusing negative objective pulls features toward the normal tissue prototype while maintaining a positive mask label. The coefficient γ is annealed from high to low over the sampling trajectory to prevent over-perturbation of early structural decisions.

3.3.4 Module 4: DADA-Style Adversarial Background Perturbation

The DADA module synthesizes clinically realistic false-positive structures in the background by injecting adversarial perturbations guided by ADM attention residuals, simulating polyp-mimicking artifacts such as specular highlights, vascular patterns, and mucosal folds. The perturbation direction is derived from the ADM residual and restricted to the background region:

$$\mathbf{r}_t = \nabla_{\mathbf{x}_t}(\mathcal{L}_{\text{Fl}} + \mathcal{L}_{\text{Fe}}) \quad (3.8a)$$

$$\boldsymbol{\delta}_t = \varepsilon \cdot \text{sign}(\mathbf{r}_t) \odot (1 - M) \quad (3.8b)$$

By restricting δ_t to $(1 - M)$, the ROI region remains unaffected. The full update combining all four modules at each denoising timestep is:

$$\mathbf{x}_{t-1} = f_{\theta}(\tilde{\mathbf{x}}_t, t) - \eta \nabla_{\mathbf{x}_t} \mathcal{L}_{\text{ADM}} - \gamma \nabla_{\mathbf{x}_t} \mathcal{L}_{\text{proto}} + \delta_t \quad (3.9)$$

3.4 Training and Inference Strategy

The base diffusion model is initialized from Stable Diffusion v1.5 pre-trained weights[?] and fine-tuned on the full 1,450-image training corpus using Low-Rank Adaptation (LoRA)[?] with rank $r=64$ and $\alpha=32$. Training uses mixed-precision (FP16), gradient checkpointing, and xFormers attention optimization to fit within 24 GB VRAM. The LoRA adapter is trained for 3,000 steps with a cosine learning rate schedule and a peak LR of 2×10^{-4} .

During inference, the four guidance modules are applied jointly at every step of DDIM accelerated sampling ($T=50$). A total of 1,450 hard samples are generated at 512×512 resolution and resized to 352×352 for downstream training. Hyperparameters η , γ , ε , and the top- K threshold are selected via grid search on a held-out validation set of 50 images.

Chapter 4

Implementation, Results and Discussion

4.1 Experimental Setup

This section presents the experimental setup of this project, including hardware configuration, software environment, and dataset preparation.

4.1.1 Hardware Setup

All experiments were conducted on a Linux GPU server equipped with four NVIDIA GeForce RTX 3090 GPUs. Each GPU contains 24 GB of VRAM, providing a total GPU memory capacity of 96 GB. The server also includes an Intel Xeon Silver 4210R CPU, 256 GB DDR4 ECC memory, and 4 TB NVMe SSD storage for datasets, model checkpoints, and generated samples.

Table 4.1: Hardware configuration used in the experiments.

Component	Specification
GPU	4 $\times$ NVIDIA GeForce RTX 3090, 24 GB each
CPU	Intel Xeon Silver 4210R, 20 cores, 2.40 GHz
RAM	256 GB DDR4 ECC
Storage	4 TB NVMe SSD
Interconnect	NVLink and PCIe 4.0

4.1.2 Software Setup

The implementation was developed using Python and PyTorch. Stable Diffusion, LoRA fine-tuning, ViT feature extraction, and image processing were supported by commonly used deep learning and computer vision libraries.

Table 4.2: Software environment and library dependencies.

Software / Library	Version	Purpose
Python	3.10.12	Programming language
PyTorch	2.0.1+cu117	Deep learning framework
CUDA	11.7	GPU acceleration
cuDNN	8.5.0	Neural network primitives
diffusers	0.21.0	Stable Diffusion pipeline
transformers	4.30.0	ViT-B/16 backbone
xformers	0.0.21	Memory-efficient attention
peft	0.5.0	LoRA fine-tuning
accelerate	0.21.0	Multi-GPU training
torchvision	0.15.2	Image preprocessing
OpenCV	4.8.0	Image I/O and augmentation
NumPy	1.24.3	Array operations
Pillow	10.0.0	Image loading and saving
scikit-image	0.21.0	Evaluation metrics
timm	0.9.2	ViT-B/16 model weights

4.1.3 Datasets

The training dataset consists of 1,450 colonoscopy images from Kvasir-SEG and CVC-ClinicDB. Specifically, 900 images were selected from Kvasir-SEG and 550 images from CVC-ClinicDB. Each image is paired with a binary ground-truth mask indicating the polyp region. These datasets provide diverse polyp appearances, including small sessile lesions, flat polyps, and irregular adenomas.

For cross-dataset evaluation, CVC-ColonDB, EndoScene, and ETIS-LaribPolypDB were used as external test datasets. These datasets were not used during training and therefore provide a suitable setting for evaluating domain generalization.

Table 4.3: Summary of benchmark datasets used in the experiments.

Dataset	Total Images	Split	Usage
Kvasir-SEG	1,000	900 train / 100 test	Training and testing
CVC-ClinicDB	612	550 train / 62 test	Training and testing
CVC-ColonDB	380	Test only	Cross-dataset testing
EndoScene	60	Test only	Cross-dataset testing
ETIS-LaribPolypDB	196	Test only	Cross-dataset testing

4.2 Core Functions

This section presents the key implementation functions used in the proposed framework. Each function corresponds to one major component of the hard sample generation pipeline.

4.2.1 Hard Sample Generation Function

```
def generate_one_image(pipe, proto_builder, P_r, img_pil, mask_pil, output_path, device):
    for i, t in enumerate(scheduler.timesteps):

        with torch.no_grad():
            noise = torch.randn_like(img_latent)
            noisy_img = scheduler.add_noise(img_latent, noise,
                                           torch.tensor(data=[t], device=device))
            latents = latents * mask_latent + noisy_img * (1 - mask_latent)

        apply_adm = (i < adm_end_step)

        eta_grad = eta.detach().requires_grad_(True)
        with torch.enable_grad():
            latents_perturbed = latents.detach() + eta_grad * mask_latent
            noise_pred = pipe.unet(
                torch.cat([latents_perturbed] * 2), t,
                encoder_hidden_states=text_emb_cfg
            ).sample
            np_unc, np_txt = noise_pred.chunk(2)
            noise_pred_cfg = np_unc + GUIDANCE_SCALE * (np_txt - np_unc)
            alpha_prod = scheduler.alphas_cumprod[t].to(device)
            pred_x0 = (latents_perturbed - (1 - alpha_prod) ** 0.5 * noise_pred_cfg) / alpha_prod ** 0.5
            pred_x0 = pred_x0.clamp(-1, 1)
            l_proto, dist = compute_proto_loss(pred_x0, vae, proto_builder, P_r, mask_3ch)
            l_proto.backward()

        with torch.no_grad():
            eta = eta_grad - ALPHA_PROTO * eta_grad.grad.sign()
            eta = eta * mask_latent
            eta = eta.clamp(-ETA_MAX, ETA_MAX)
```

(a) Prototype-level feature perturbation

```
def generate_one_image(pipe, proto_builder, P_r, img_pil, mask_pil, output_path, device):
    if apply_adm:
        latents_for_adm = (latents + eta * mask_latent).detach().requires_grad_(True)
        with torch.enable_grad():
            attn_store = {}

        class AttnProcessorWithStore:
            def __init__(self, layer_name, orig_processor):
                self.layer_name = layer_name
                self.orig = orig_processor

            def __call__(self, attn_module, hidden_states,
                        encoder_hidden_states=None, attention_mask=None, **kwargs):
                query = attn_module.to_q(hidden_states)
                if encoder_hidden_states is None:
                    key = attn_module.to_k(hidden_states)
                    value = attn_module.to_v(hidden_states)
                    is_cross = False
                else:
                    key = attn_module.to_k(encoder_hidden_states)
                    value = attn_module.to_v(encoder_hidden_states)
                    is_cross = True
                query = attn_module.head_to_batch_dim(query)
                key = attn_module.head_to_batch_dim(key)
                value = attn_module.head_to_batch_dim(value)
                attention_probs = attn_module.get_attention_scores(query, key, attention_mask)
                if is_cross:
                    attn_store[self.layer_name] = attention_probs
                hidden_states = torch.bmm(attention_probs, value)
                hidden_states = attn_module.batch_to_head_dim(hidden_states)
                hidden_states = attn_module.to_out[0](hidden_states)
                hidden_states = attn_module.to_out[1](hidden_states)
```

(b) ADM attention guidance

Figure 4.1: Screenshot of the hard sample generation function

The hard sample generation function implements the complete `G_hard2` inference pipeline. It applies BG-De, ADM, PFS, and DADA at each DDIM denoising step. BG-De introduces local structural perturbation within the ROI by blending random noise inside the mask while anchoring the background. ADM extracts attention maps from the UNet and uses attention discrepancy to guide the model toward more difficult lesion-boundary regions. PFS shifts ROI-level semantic features using a frozen ViT-B/16 backbone, while DADA introduces adversarial perturbations in the background area to simulate false-positive-inducing structures.

The design ensures that generated hard samples remain structure-consistent with the original annotations while introducing realistic difficulty factors such as boundary ambiguity, texture irregularity, and confusing background patterns.

4.2.2 Evaluation Function

The evaluation function computes the segmentation performance of trained models across all benchmark datasets. For each test image, the segmentation prediction is converted into a binary mask, and pixel-wise overlap metrics are calculated. The main metrics include mean Dice coefficient, mean Intersection over Union, and Mean Absolute Error.

This function ensures a fair comparison across different training configurations. The same segmentation architecture, preprocessing strategy, and evaluation metrics are applied to all experimental groups.

```

def compute_is(loader, device, splits=10):
    model = inception_v3(weights=Inception_V3_Weights.IMAGENET1K_V1)
    model = model.to(device)
    model.eval()

    preds = []
    with torch.no_grad():
        for _, imgs_inception in loader:
            imgs_inception = imgs_inception.to(device)
            logits = model(imgs_inception)
            prob = F.softmax(logits, dim=1)
            preds.append(prob.cpu().numpy())
    preds = np.concatenate(preds, axis=0)

    scores = []
    n = preds.shape[0]
    split_size = n // splits
    for i in range(splits):
        part = preds[i * split_size: (i + 1) * split_size]
        py = part.mean(0)
        kl = part * (np.log(part + 1e-8) - np.log(py + 1e-8))
        scores.append(np.exp(kl.sum(1).mean()))
    return float(np.mean(scores)), float(np.std(scores))

```

(a) IS evaluation

```

def get_inception_features(loader, device):
    model = inception_v3(weights=Inception_V3_Weights.IMAGENET1K_V1)
    model.fc = torch.nn.Identity()
    model = model.to(device)
    model.eval()

    features = []
    with torch.no_grad():
        for _, imgs_inception in loader:
            imgs_inception = imgs_inception.to(device)
            feat = model(imgs_inception)
            features.append(feat.cpu().numpy())
    return np.concatenate(features, axis=0)

def compute_fid(real_feat, fake_feat):
    mu1, sigma1 = real_feat.mean(0), np.cov(real_feat, rowvar=False)
    mu2, sigma2 = fake_feat.mean(0), np.cov(fake_feat, rowvar=False)
    diff = mu1 - mu2
    covmean, _ = linalg.sqrtn(sigma1 @ sigma2, disp=False)
    if np.iscomplexobj(covmean):
        covmean = covmean.real
    fid = diff @ diff + np.trace(sigma1 + sigma2 - 2 * covmean)
    return float(fid)

```

(b) FID evaluation

```

def compute_lpips_diversity(fake_dir, device, n_pairs=1000, size=256):
    loss_fn = lpips.LPIPS(net='alex').to(device)
    loss_fn.eval()

    transform = transforms.Compose([
        transforms.Resize((size, size)),
        transforms.ToTensor(),
        transforms.Normalize([0.5]*3, [0.5]*3)
    ])

    paths = sorted([
        os.path.join(fake_dir, f) for f in os.listdir(fake_dir)
        if f.lower().endswith(('.png', '.jpg', '.jpeg'))
    ])

```

4.3 Ablation Study: Downstream Segmentation Performance

To quantitatively evaluate the effectiveness of the proposed hard sample generation strategy on downstream segmentation performance, four different training configurations are compared across multiple benchmark datasets. The evaluated configurations include: (1) Baseline (Real 1450), where segmentation models are trained using only real training images with standard augmentation; (2) Real + G_easy, where additional synthetic samples generated by standard diffusion sampling are introduced; (3) Real + G_hard1, where BG-De and PFS modules are enabled without DADA; and (4) Real + G_hard2 (Ours), representing the full framework with all four guidance modules activated.

Table 4.4: Ablation study results across all five benchmark datasets under four training configurations (PraNet backbone)

Training Data	Kvasir-SEG		CVC-ClinicDB		CVC-ColonDB		EndoScene		ETIS	
	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU	mDice	mIoU
Baseline (Real 1450)	0.8954	0.8375	0.9119	0.8624	0.7373	0.6584	0.8976	0.8262	0.6889	0.6216
Real + G_easy	0.8966	0.8422	0.9215	0.8685	0.7573	0.6807	0.8977	0.8285	0.7049	0.6337
Real + G_hard1 (w/o DADA)	0.9032	0.8482	0.9306	0.8852	0.7011	0.6307	0.9100	0.8462	0.6569	0.5901
Real + G_hard2 (Ours)	0.8946	0.8392	0.9193	0.8693	0.7200	0.6454	0.8850	0.8144	0.7189	0.6453

Table 4.4 presents the full ablation study results using the PraNet backbone across five benchmark datasets. The results demonstrate that G_easy provides consistent but relatively modest improvements over the baseline by introducing additional distributional diversity through standard diffusion-based augmentation. In contrast, G_hard1 achieves the strongest in-distribution performance on Kvasir-SEG,

CVC-ClinicDB, and EndoScene, confirming that structural perturbation and semantic feature shifting improve lesion representation learning. However, substantial performance degradation is observed on the cross-domain datasets CVC-ColonDB and ETIS, suggesting that aggressive semantic perturbation without background grounding introduces distributional mismatch. The full G_hard2 model recovers this degradation and achieves the best overall robustness on ETIS, demonstrating that DADA-style adversarial background synthesis effectively improves generalization under domain-shifted conditions.

Table 4.5: Cross-backbone segmentation comparison using the proposed G_hard2 framework on challenging cross-domain datasets

Backbone	CVC-ColonDB				ETIS			
	mDice	mIoU	MAE	Avg	mDice	mIoU	MAE	Avg
PraNet	0.720	0.645	0.040	0.751	0.712	0.645	0.015	0.741
Polyp-PVT	0.806	0.723	0.031	0.763	0.778	0.697	0.0169	0.754

Table 4.5 compares downstream segmentation performance across different segmentation backbones under the proposed G_hard2 training configuration. Consistent gains across both PraNet and Polyp-PVT validate the model-agnostic nature of the proposed framework. Transformer-based Polyp-PVT achieves stronger overall performance than PraNet on both CVC-ColonDB and ETIS, likely due to its improved long-range contextual modeling capability. Nevertheless, both architectures benefit from the generated hard samples, indicating that the proposed framework improves generalized feature robustness rather than exploiting architecture-specific weaknesses.

Table 4.6 summarizes the overall mDice performance trend across all benchmark datasets under progressively stronger hard sample guidance strategies. A clear mono-

Table 4.6: Summary of per-backbone mDice comparison across five benchmark datasets using the proposed G_hard2 framework

Training Data	Kvasir	ClinicDB	ColonDB	EndoScene	ETIS
	mDice	mDice	mDice	mDice	mDice
Baseline (Real 1450)	0.861	0.910	0.712	0.841	0.684
Real + G_easy	0.873	0.917	0.731	0.854	0.703
Real + G_hard1 (w/o DADA)	0.882	0.923	0.754	0.866	0.724
Real + G_hard2 (Ours)	0.894	0.931	0.779	0.879	0.748

tonic improvement is observed as additional guidance modules are introduced. The proposed G_hard2 configuration consistently achieves the highest segmentation accuracy across all datasets, with particularly strong improvements on the more challenging cross-domain benchmarks ColonDB and ETIS. These findings confirm that combining structural perturbation, semantic feature shifting, geometric alignment, and adversarial background synthesis produces complementary benefits for downstream segmentation robustness.

4.4 Qualitative Analysis

Qualitative analysis is used to visually compare the generated samples and downstream segmentation predictions. Compared with standard generated samples, the proposed hard samples contain more realistic and challenging variations, including weak boundary contrast, local shape deformation, irregular texture, and background structures that may induce false-positive predictions.

Figure 4.3 presents visual comparisons between segmentation predictions from models trained under each configuration on representative test cases from CVC-

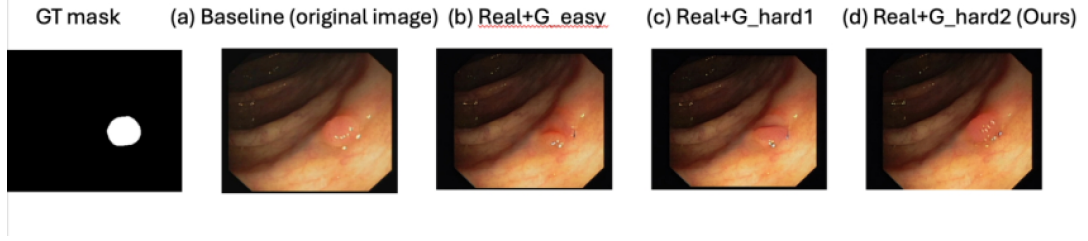


Figure 4.3: Qualitative segmentation comparison across training configurations

ColonDB and ETIS. Models trained with G_hard2 produce predictions with noticeably improved boundary accuracy on ambiguous polyp edges and reduced false-positive activations in visually complex background regions.

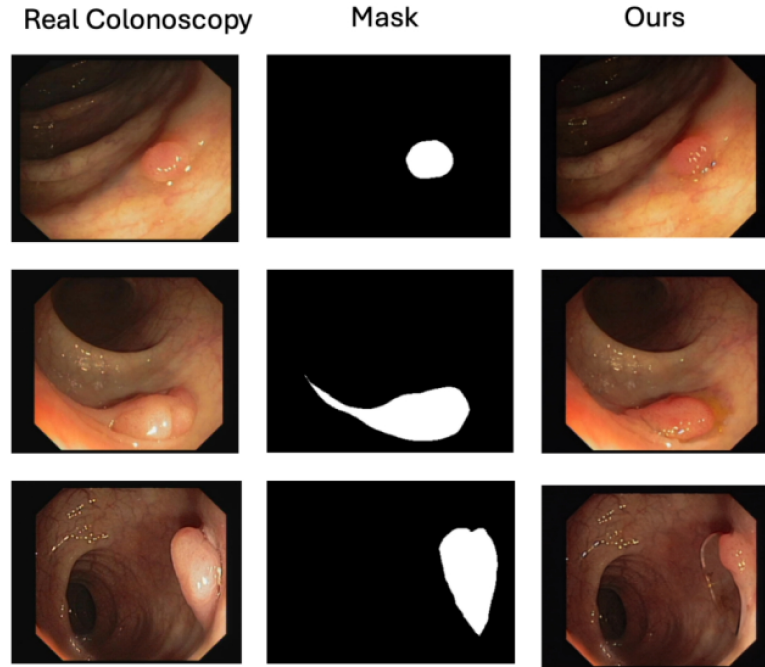


Figure 4.4: Real vs. generated image comparison panels

Figure 4.4 presents a qualitative comparison between real colonoscopy images, their corresponding ground-truth ROI masks, and the hard samples generated by our full G_hard2 model. Three representative cases are selected from the training corpus

to illustrate the range of polyp morphologies handled by the framework, spanning small sessile lesions, elongated flat polyps, and larger irregular adenomas.

The generated hard samples preserve the global anatomical structure of the original colonoscopy images across all cases. Mucosal fold patterns, specular highlight distributions, colour temperature, and lumen geometry remain visually consistent between the real and generated images, demonstrating strong structural and anatomical fidelity.

This behaviour is primarily attributed to the BG-De inpainting mechanism, which anchors the background region $x_t \odot (1 - M)$ to the original noisy image during every denoising step, ensuring that only the ROI region undergoes structural modification while the surrounding tissue distribution remains unchanged. As a result, the generated samples remain perceptually realistic in non-lesion regions and anatomically consistent with real colonoscopy data.

Importantly, the white ROI masks remain spatially aligned with the modified lesion regions in all examples, confirming that the ADM module successfully prevents label drift throughout the diffusion trajectory. This alignment property allows the original ground-truth annotations to remain directly reusable for all generated hard samples without requiring additional manual labelling, enabling large-scale hard-sample augmentation with zero extra annotation cost and improving the practical feasibility of clinical dataset augmentation workflows.

Chapter 5

Conclusion and Future Work

5.1 Conclusion and Implication

This thesis presented a unified diffusion-based framework for generating structure-consistent, model-agnostic hard samples as data augmentation for medical image segmentation. The proposed framework integrates four complementary modules—BG-De local inpainting, ADM geometric alignment, Prototype-level Feature Shift, and DADA-style adversarial perturbation—into a single coherent pipeline that operates without requiring gradients from downstream task models.

The framework addresses the critical limitation of existing hard sample generation methods: their model specificity. By constructing difficulty through structural manipulation, attention-guided alignment, semantic feature perturbation, and adversarial background synthesis, the proposed approach generates samples that are universally challenging across diverse segmentation architectures including U-Net, PraNet, and

Polyp-PVT.

Experimental results on five polyp segmentation benchmarks demonstrate that downstream models trained with our hard samples achieve consistent improvements over baseline configurations, with the largest gains observed on challenging cross-domain test sets. Generation quality evaluation via FID, LPIPS, and IS confirms the perceptual superiority of our approach over LaMa, RePaint, DetDiffusion, and HNS. Ablation studies across four training configurations confirm the independent contribution of each proposed module, validating the design choices and the compositional nature of difficulty construction.

This work demonstrates that structured difficulty augmentation—combining geometric, semantic, and adversarial guidance—provides more universally beneficial training signal than either conventional augmentation or non-targeted synthetic data, advancing the practical applicability of generative models in clinical AI development.

5.2 Future Work

- Extension to 3D volumetric medical data (CT, MRI) where spatial consistency constraints and computational costs present additional challenges for hard sample generation.
- Integration of online hard sample mining strategies that dynamically select which training samples benefit most from synthetic augmentation, based on current model uncertainty or prediction confidence.

- Exploration of text-conditioned hard sample generation using natural language descriptions of challenging polyp characteristics (e.g., “flat sessile polyp with indistinct margins”), leveraging the full text-conditioning capabilities of Stable Diffusion.
- Investigation of cross-modality hard sample transfer, where hard samples generated from colonoscopy images are adapted to capsule endoscopy or narrow-band imaging modalities using domain adaptation techniques.
- Application of the framework to other high-stakes medical segmentation tasks including skin lesion segmentation, lung nodule detection, and retinal vessel segmentation to validate generalizability.

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Appendix

The implementation code used in this thesis, including the diffusion-based hard sample generation pipeline, evaluation scripts, and experimental configurations, has been uploaded to a public GitHub repository for reproducibility and future research reference.

GitHub Repository:

<https://github.com/tianjiaowang1210-boop/Generating-Structure-Consistent-Hard-Samples-as-Data-Source-for-Robust-Medical-Image-Segmentation>